

XIAYU (SUMMER) ZHAO

PhD Student — Robotic, Automation, and Intelligent Sensing (RAISe) Lab
University of Illinois Urbana-Champaign (UIUC)
Civil and Environmental Engineering – The Grainger College of Engineering
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Research Interests

Construction robotics; aerial-ground multi-robot systems (UAV-UGV, hexapod, and soft pneumatic platforms) for contact-based infrastructure inspection; visual-tactile-spatial multimodal sensing; adaptive robotic control and reinforcement learning; LLM-driven robot navigation and human-robot collaboration; computer vision for defect inspection.

Education

Ph.D. Student, *Civil and Environmental Engineering, The Grainger College of Engineering, University of Illinois Urbana-Champaign* Spring 2024 – present

- Advisor: Dr. Houtan Jebelli
- GPA: 3.91/4.00 (Spring 2024 – present)
- Dissertation Title:

Master of Computer Science, *Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign* 2025 – 2027

- Research fields: Vision, Graphics, Interaction, and Robotics.
- GPA: 4.00/4.00 (Fall 2025 – present)
- Courses taken: CS598 (Deep Learning for Robotic Manipulation) A, CS547/IE534 (Deep Learning) A+, ECE470 (Introduction to Robotics) A, CS598 (Advanced Computational Topics in Robotics) A-, CS443 (Reinforcement Learning) B+, Ordered Data Structures (Coursera, provided by UIUC), Object-Oriented Data Structures in C++ (Coursera, provided by UIUC).

Master of Architecture, *Tsinghua University, Beijing, China* 2019 – 2021

- Research fields: Robotic Construction, 3D Printing, Computational Design, GPA: 3.97/4.00

Bachelor of Architecture, *Tianjin University, Tianjin, China* 2014 – 2019

- Research fields: Computational Design, GPA: 3.85/4.00

Academic Appointments

Graduate Student Research Assistant, *University of Illinois Urbana-Champaign, Urbana, IL, U.S.*
Spring 2024 – present

Supervisor: Professor Houtan Jebelli

Graduate Student, *Icon Lab, Penn State University* Fall 2023

Robotic Construction Safety Training Prototype with Immersive Technology (VR)

Role: **Team Leader** — Supervisor: Professor John Messner — TA: Saleh Alghamdi

Team: Tessa Beauchat, Islam Gharaibeh, Fangxiao Li

- Connected ROS (Robot Operating System) with Unity 3D to control robot arm mounted UGV (Unmanned Ground Vehicle) in immersive simulation environments.
- Developed an interactive, game-based prototype specifically for collision-avoidance safety training, targeted at workers involved in human-robot collaborative construction scenarios.

Research Assistant, *DAIC (Digital Architecture and Robotic Construction) Lab, Tsinghua University*
Jul. 2021 – Apr. 2023

Supervisor: Professor Weiguo Xu

- Extracted design information to program and control robot arms for large-scale 3D printing.

- Developed software plug-ins for robotic construction, including 3D printing, metal cutting, wood cutting, wall painting, etc.

Master's Thesis and Graduation Design

Spring 2021

Large-Scale Multi-Robots 3D Printing Construction Planning — Instructor: Professor Weiguo Xu

Student Assistant, *Built Environmental Virtual Reality Lab, Tianjin University*

2016 – 2017

Instructor: Professor Sinan Yuan

Sponsor: *National Natural Science Foundation*

- Related Methods: Field research, Data Collection, 3D Modeling, Data Input, Data Visualization using a Game Engine, VR Experience for Volunteers, Analysis of Statistical Results, Spatial morphology, and Human Spatial Cognition Mechanisms.

Teaching and Mentoring

Teaching Assistant, *Robotic Construction Research Series, Robotic Construction Group, Tsinghua University*

Sep. 2022 – Apr. 2023

Supervisor: Professor Weiguo Xu

- Advising master students with their graduation projects related to robotic construction.
- Research topics involved plastic 3D printing, wood cutting, and foam cutting with KUKA robot arm.

Teaching Assistant, *Tsinghua University Parametric Design Workshop 2019*

Jul. 2019

Tutor: Casey Rehm

Project Title: *Artificial Intelligence Architectural Design – 2D and 3D Generative Neural Network Techniques for Design Biomorphism.*

Teaching Assistant, *Computer-Aided Architectural Design Research in Asia (CAADRIA) Workshop 2018*

May 2018

Tutors: Shajay Bhooshan, Vishu Bhooshan, Henry Louth, Leo Claudius

Project Title: *Developable Funicular Skeletons with Tensioning Membranes*

Professional Experiences

Research Engineer, *Institute of Future Human Habitats*

2021 – 2023

Tsinghua University Shenzhen International Graduate School (SIGS)

- Developed large-scale robotic construction technology based on composite consumables and practical building materials with an interdisciplinary team.
- Developed software interface to streamline input and control.

3D Concrete Printing Specialist, *Heqing Digital Building Technology Co.*

2020 – Apr. 2023

- Participated in several on-site robotic construction projects in China and the US, including designing the shape, designing the print path, writing the print program, operating the robot arm on-site, post-maintenance, etc.

Intern Architect, *Zaha Hadid Architects*

Jun. – Sept. 2020

Project Director: Satoshi Ohashi

- Participated in the design stage of the final winning bid – competition – design of *Tower C of Shenzhen Bay Super Headquarters.*
- Participated in the Preliminary design of the Unicorn Island project in Sichuan.

Co-designer, *Model Slicing and Toolpath Plug-in Development for Robotic Construction*

Mar. – Apr. 2023

Project Leaders: Chenwei Sun, Weiguo Xu

Principal Designer, *Rotatable Fine Extrusion Print Head Design for 3DCP Decorative Panels*

Oct. 2021 – Feb. 2022

Partners: Zilu Guo, Qi Liu, Chenwei Sun

Associate Project Manager, *Wooden House Renovation with Multi-Robots 3D Printed Walls* Jul. – Aug. 2022

Project Leaders: Weiguo Xu, Chenwei Sun

Sponsor and Co-designer: University of Hong Kong, Prof. Lin John C.H.

Principal Designer, *Factory Façade Design Based on Style Transfer Machine Learning* 2022
Zhangjiakou, Hebei, China

Lead Designer, *Robot 3D Concrete Printed Model House for Low-income People in Africa* 2020
Project Leader: Weiguo Xu — Wuxi, Zhejiang, China

Designer and Engineer, *Wujiashuang Multi-Robots 3D Concrete Printed Farmhouse* 2020
Project Leaders: Weiguo Xu, Yuan Gao, Chenwei Sun — Wujiashuang, Hebei, China

Awards & Honors

Academic Awards

- **Olympiad Medalist** – All Engineering Field, 2025 Olympiad in Engineering Science 2025
Held at the University of Stavanger, Norway
Submission: *Autonomous Drone-based System for Precision, Non-contact Surface Finishing in Construction*
Team: Tianyu Ren, Xiayu (Summer) Zhao, Houtan Jebelli
- **First Place** – ASCE 2025 Global Robotics and Automation Competition 2025
2025 ASCE Computing Conference, New Orleans, hosted by the ASCE Computing Division
Submission: *Module-enhanced Slope-Adaptive and Hazard-Aware Hexapod Robotic System for Safe Roof Inspection*
Team: Xiayu (Summer) Zhao, Houtan Jebelli
- **Journal Cover Feature** – *Computer-Aided Civil and Infrastructure Engineering* 2025
Paper: *A Computational Method for Real-time Roof Defect Segmentation in Robotic Inspection*
- **Outstanding Graduate Award** 2019
- **Outstanding Graduation Design Award** 2019
- **Tianjin University's Outstanding Student Award** (1/1000) 2019
- **Outstanding Project Assignment in National Higher Education Institutions** 2017
- **Honorable Mention** – Skyscraper Competition (eVolo) 2018
Team: Jiangchen Hu, Yining Bei, Mingwen Zhang
- **China Architecture Newcomers Competition – Top 100 Award** 2017
- **4th National Tournament of Green Building Innovation (2015) – First Prize** 2015
Team: Adela Wei, Mark Wang, Mason Yang, Yu Zhao
Sponsor: *The Chartered Institute of Building (CIOB)*

Academic Scholarship

- **University Graduate Fellowship** (Sponsor: *Pennsylvania State University*) 2023
- **College of Engineering Scholarship** (Sponsor: *College of Engineering, PSU*) 2023
- **China National Scholarship** (1/101) 2018
- **China National Scholarship** (1/97) 2016
- **China National Scholarship** (1/94) 2015
(Sponsor: *Ministry of Education of the People's Republic of China*)

Patent & Software Copyright

All applications are actively processed by the University of Illinois Office of Technology Management (OTM) for United States Provisional Application.

Patent Applications

1. BEAM (Bidirectional Environment Abstraction for Multi-Robot): A Method and System for Robotic Navigation in Construction Environments

2. Modular Bi-Directionally Bendable Soft Pneumatic Multi-Leg UAV Landing and Perception Platform for Contact-Based Infrastructure Inspection
3. Soft Pneumatic UAV Landing and Perception Platform with Visual-Tactile-Spatial Multimodal Sensing for Contact-Based Infrastructure Inspection
4. UAV-Deployable Soft-Rigid Hexapod Ground Module with Auto-Spinning Locking Undercarriage and Flippable Multimodal Inspection Arm for Non-Destructive Inspection and Multimodal Data Collection on Fragile Infrastructure

Software Copyright Applications

1. BEAM (Bidirectional Environment Abstraction for Multi-Robot): A Software Framework for Robotic Navigation in Construction Environments
2. SaHa: Slope-Adaptive and Hazard-Aware Robotic Control Software for Hexapod Roof Inspection
3. A UAV-Deployable Soft-Rigid Hexapod Ground Module for Multimodal Visual-Tactile-Spatial Data Collection on Fragile Infrastructure

Publications

Publication records checked against Google Scholar, September 2026.

Journal Articles

1. **Zhao, X.**, Liu, Y., Jebelli, H. (2026). “A soft pneumatic UAV landing and perception platform for gentle contact-based infrastructure inspection via visual-tactile-spatial multimodal sensing.” *Computer-Aided Civil and Infrastructure Engineering*, 100104. [Paper](#).
2. **Zhao, X.**, Liu, Y., Jebelli, H. (2025). “Module-Enhanced Slope-Adaptive and Hazard-Aware Hexapod Robotic System for Safe Roof Inspection.” *ASCE OPEN: Multidisciplinary Journal of Civil Engineering*, 3(1), 04025011. [Paper](#).
3. **Zhao, X.**, Jebelli, H. (2025). “A computational method for real-time roof defect segmentation in robotic inspection.” *Computer-Aided Civil and Infrastructure Engineering*, 40(23), 3596-3623. [Paper](#).
4. Ye, W., Chen, S., **Zhao, X.**, Xu, W. (2022). “Porous space—biomimetic of tafoni in computational design.” *Architectural Intelligence*, 1(1), 18. [Paper](#).
5. **Zhao, X.**, Xu, W. (2021). “Research Progress and Application Status of 3D Printing Construction Technology.” *Chinese & Overseas Architecture*, 10. [Paper](#).
6. **Zhao, X.**, Xu, W. (2021). “Research Progress and Current Applications of 3D Printing Construction Technology.” *Chinese & Overseas Architecture*, 10, 7-13. [Paper](#).

Conference Papers

7. Guo, X., Li, Y., Kong, X., Jiang, Y., **Zhao, X.**, Gong, Z., Zhang, Y., Li, D., et al. (2026). “Toward engineering AGI: Benchmarking the engineering design capabilities of LLMs.” *Advances in Neural Information Processing Systems*, 38. [Paper](#).
8. **Zhao, X.**, Liu, Y., Driggs-Campbell, K., Jebelli, H. (2026). “Tactile-Aware Soft Hexapod UAV Gripper for Multi-Modal Aerial Manipulation in Construction.” *Construction Research Congress 2026*, 196-206. [Paper](#).
9. **Zhao, X.**, Jebelli, H. (2026). “Open-Vocabulary, Context-Aware Robot Navigation on Construction Sites: Integrating LLM-Driven Value Map Composition with Hierarchical Scene Graphs.” *Construction Research Congress 2026*, 227-237. [Paper](#).
10. **Zhao, X.**, Ren, T., Jebelli, H. (2026). “LLM-Driven Safe Robotic Navigation in Construction Sites: A Prompt Engineering Framework with Terrain-Adaptive Path Planning.” *Computing in Civil Engineering 2025: Computational and Intelligent Technologies*, 57-67. [Paper](#).

11. **Zhao, X.**, Liu, Y., Jebelli, H. (2026). “Hybrid UAV-UGV System with Interchangeable Ground Modules for Multi-Terrain Construction Site Inspection.” *Computing in Civil Engineering 2025: Resilient, Robotic, and Educational Systems*, 462-472. [Paper](#).
12. Ren, T., **Zhao, X.**, Jebelli, H. (2026). “Dynamic Robotic Control Strategies for UAV Stability with Rotatory Payload.” *Computing in Civil Engineering 2025: Resilient, Robotic, and Educational Systems*, 452-461. [Paper](#).
13. Ren, T., Hasan, I., **Zhao, X.**, Liu, Y., Jebelli, H. (2026). “ConstrucSim: A Reproducible Platform for Dexterous Humanoid Robot Perception, Planning, and Contact in Construction.” *Construction Research Congress 2026*, 45-55. [Paper](#).
14. Ren, T., **Zhao, X.**, Jebelli, H. (2025). “Advanced Sensor Integration for Enhanced Flight Control in UAV-Based Construction Automation.” *Proceedings of the 42nd International Symposium on Automation and Robotics in Construction (ISARC)*, Montreal, QC, Canada, 28-31. [Paper](#).
15. Ren, T., **Zhao, X.**, Jebelli, H. (2025). “Autonomous Drone-Based System for Precision, Non-contact Surface Finishing in Construction.” *Olympiad in Engineering Science*, 580-591. [Paper](#).
16. **Zhao, X.**, Ren, T., Liu, Y., Jebelli, H. (2025). “A Pneumatic Sensor-Equipped Hexapod Landing Gear and Gripper System for Gentle Drone Contact with Building Envelopes and Energy Assets.” *Olympiad in Engineering Science*, 592-609. [Paper](#).
17. **Zhao, X.**, Ren, T., Liu, Y., Jebelli, H. (2025). “Sensor Fusion and Hierarchical Inspection Strategy for Aerial-Ground Multi-robot System.” *42nd International Symposium on Automation and Robotics in Construction (ISARC 2025)*. [Paper](#).
18. Liu, J., Ping, Y., Wang, Y., **Zhao, X.** (2025). “Machine Learning-Based Optimization for Modular Robotic Systems in Construction: A Review.” *Computing in Civil Engineering 2024*, 851-860. [Paper](#).
19. **Zhao, X.**, Jebelli, H. (2025). “Adaptive kinematic and dynamic control for hexapod robots in complex construction environments.” *Proc., 42nd Int. Symp. on Automation and Robotics in Construction. Montreal, Canada: Concordia University*. [Paper](#).
20. Ojha, A., Gautam, Y., **Zhao, X.**, Sharifironizi, A., Jebelli, H., Martin, A. (2024). “Understanding the Impact of an Exoskeleton System on the Metabolic Cost of Workers during Common Repetitive Construction Tasks.” *Computing in Civil Engineering 2024*, 646-655. [Paper](#).
21. Ojha, A., **Zhao, X.**, Zhang, Y., Liu, Y., Jebelli, H. (2024). “Improving Heat Strain Prediction in Construction Workers: A Transfer Learning Approach to Overcome Variability Challenges.” *Computing in Civil Engineering 2024*, 877-886. [Paper](#).
22. **Zhao, X.**, Jebelli, H. (2024). “Grounding Large Language Models in Robot Control: Facilitating Human-Robot Collaboration in Construction.” *Computing in Civil Engineering 2024*, 933-941. [Paper](#).
23. **Zhao, X.**, Ren, T., Jebelli, H. (2024). “Adaptive Intelligence for Robot Navigation Efficiency with a Deep Reinforcement Learning-Based Cyber-Physical System.” *Computing in Civil Engineering 2024*, 831-840. [Paper](#).
24. Ye, W., **Zhao, X.**, Xu, W. (2022). “Simulation Algorithm Based on Weathered Rock Morphology and Optimization Algorithm for Design Applications.” *The International Conference on Computational Design and Robotic Fabrication*, 110-121. [Paper](#).

Additional Conference Abstract

- **Zhao, X.**, Xu, Z., Xu, W. (2023). “Reusable Assembled Lightweight 3D Concrete Printed Components Based on Tessellation Patterns.” *IASS Symposium 2023*. (Abstract Accepted).

Invited Interviews, Exhibitions and Speeches

Exhibitions and Invited Speeches Bi-City Biennale of Urbanism/Architecture 2022 Apr. 2022 – Jan. 2023

Title: *Assembled 3D Printed Concrete Urban Miniature Garden*

Role: **Lead Exhibit Designer** — *Kingway Brewery*, Shenzhen, Guangdong, China

Design as Solution: Shenzhen Design Week 2021

Dec. 2021

Title: *Robotic 3D Printed Concrete Construction Works*

Role: **Keynote Speaker & Lead Exhibit Designer** — *UpperHills*, Shenzhen, Guangdong, China

Future-Home: Beijing Urban Architecture Biennale 2021

Oct. 2021

Role: **Exhibit Designer** — *Zhangjiawan Design Town*, Beijing Urban Vice Center

Invited Interviews *To be updated.*

Professional Leadership & Service Activities

Journal & Conference Reviewer (Selected) *Journal Reviewer*

- Automation in Construction
- Engineering Applications of Artificial Intelligence
- Developments in the Built Environment
- Journal of Mechanical Engineering Science
- ASCE Open Journal

Conference Reviewer

- ASCE International Conference on Computing in Civil Engineering (i3CE 2024)
- ASCE International Conference on Computing in Civil Engineering (i3CE 2025)
- ASCE International Conference on Computing in Civil Engineering (i3CE 2026)
- 42nd International Symposium on Automation and Robotics in Construction (ISARC 2025)
- 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026)
- 12th International Conference on Industrial Ecology (ISIE 2025)
- ASCE CI & CRC 2026 Joint Conference
- 2026 IEEE International Conference on Intelligence and Safety for Robotics / Safety of Industrial Automated Systems (ISR-SIAS 2026)

Student Member, American Society of Civil Engineers (ASCE)

Fall 2023 – present

Cat Foster Volunteer, Vermilion County Animal Shelter

2025 – present

Team Leader, Data Forum Center, Tsinghua University Student Union

Fall 2020

Job Contents: Campus Applet Development; Data Visualization.

Activity Manager, Sports Department of Graduate Student Union, Tsinghua University

Fall 2019

Event Organizer, Purple (LGBTQ+ Community), Tsinghua University

Fall 2019 – Spring 2021

Team Member, Women's Soccer Team, School of Architecture, Tsinghua University

2019 – 2020

Computer Skills

Robot Programming/OS: ROS, Industrial Robot Language (URScript), C#

Programming: Matlab, Python, R, C++

Simulation Software: Unity, Unreal Engine

Statistical Analysis Software: IBM-SPSS

Drawing and Building Information Modeling: AutoCAD, Revit, Rhino, Grasshopper, Sketch-Up, SolidWorks, Adobe Photoshop, Adobe Illustrator

General Software/Platform: MS-Office, L^AT_EX, Google Colab, Jupyter Notebook

References

Available upon request.